College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 07

Focus: Integrated class design, polymorphism, validation & exceptions

Difficulty: Moderate-Hard

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Design a mini inventory.

a. Specify fields and methods for `Item` (code, name, qty) with encapsulation rules. **(6 marks)**

b. Implement the class with constructor, getters, `restock`, and `issue` (issue must not make qty negative). **(10 marks)**

c. Explain how you would unit-test `issue` manually from `main` with two scenarios. **(4 marks)**

Question 2 [20 Marks]

Inheritance + overriding under rules.

a. Create `Ticket` with method `price()` returning a base amount. **(4 marks)**

b. Create `StudentTicket` and `VipTicket` with different price rules via override. **(10 marks)**

c. Demonstrate polymorphic pricing in a loop over a `Ticket[]`. **(6 marks)**

Question 3 [20 Marks]

throw for domain rules.

a. Rewrite a setter or withdraw method so invalid input throws `IllegalArgumentException`. **(8 marks)**

b. Catch the exception in `main` and keep a menu running. **(8 marks)**

c. Compare throwing vs printing an error message inside the class - when is throwing better? **(4 marks)**

Question 4 [20 Marks]

```
class Node {
    int value;
    Node next;
}
public class Trace {
    public static void main(String[] args) {
        Node a = new Node();
        a.value = 1;
        Node b = a;
        b.value = 2;
        System.out.println(a.value);
        b = new Node();
        b.value = 3;
        System.out.println(a.value);
    }
}
```

Trace and fix.

a. What is printed by each `println`? Explain with stack/heap reasoning. **(10 marks)**

b. Draw (or describe) references after `b = new Node()`. **(6 marks)**

c. What would `a.next.value` do if `a.next` was never set? **(4 marks)**

Question 5 [20 Marks]

Abstract reporting.

a. Design abstract `Report` with abstract `generate()` and concrete `header()`. **(8 marks)**

b. Implement `SalesReport` and `AttendanceReport`. **(8 marks)**

c. Show how a method can accept `Report r` and call `generate` without knowing the concrete type. **(4 marks)**

Question 6 [20 Marks]

Composition deep dive.

a. Model a `Computer` composed of `Cpu` and `Memory` (composition). Write constructors that create the parts inside `Computer`. **(10 marks)**

b. Explain why deleting/losing the `Computer` reference means the parts are no longer needed by the program. **(4 marks)**

c. How would the design change if `Memory` could be swapped and reused in another computer (aggregation)? **(6 marks)**

Question 7 [20 Marks]

Safe console menu (integration).

a. Write a helper `readInt(Scanner sc)` that loops until a valid integer is entered (use try/catch). **(8 marks)**

b. Build a 3-option menu that uses the helper and exits cleanly on option 0. **(8 marks)**

c. State two user mistakes your menu now survives. **(4 marks)**

******* END OF QUESTION PAPER *******